PRESENTING THE HOPF ALGEBRA OF G AS A MONOGENIC EXTENSION OF THE HOPF ALGEBRA OF H

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. It opens by reducing to the preceding discussion of the same section (the discussion being a standing set-up of the section rather than a labelled statement), and then verifies the substitution explicitly; nothing further is suppressed.

Editorial note. The label lemma-monogenic-one occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The counted spans contain no label reference and no bibliography citation, so the card carries no navigation stub and no bibliography excerpt.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The running situation of the section: the group scheme G, its subgroup H and the element Phi(s). Same chapter, the preceding discussion in which the polynomial relation Phi(s) = t and the Hopf algebras of G and H are introduced. Supplies the objects and the polynomial relation on which the whole computation rests.
- Monogenic extension of a Hopf algebra. Same chapter, the notion of a monogenic extension of a Hopf algebra, i.e. one generated by a single element. Supplies the second assertion of the statement, which is read off from the displayed presentation.
- Localization A[s, 1/(pi s + 1)] and its presentation by a quotient. Same chapter together with the ambient algebra conventions: adjoining one variable and inverting an element, and the identification of the resulting algebra with a quotient by the relation Phi(s) - t. Supplies the isomorphism displayed in the statement and the rewriting of 1/(pi^ell t + 1) carried out in the proof.

Lemma 0.1. *We have*

$$A[s, \frac{1}{\pi s + 1}] = \left(A[t, \frac{1}{\pi^\ell t + 1}] \right) [s]/(\Phi(s) - t)$$

In particular, the Hopf algebra of G is a monogenic extension of the Hopf algebra of H.

Proof. Follows from the discussion above and the shape of $\Phi(s)$. In particular, note that using $\Phi(s) = t$ the element $\frac{1}{\pi^\ell t+1}$ becomes the element $\frac{1}{(\pi s+1)^\ell}$. □

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